

A_m : has Weyl weight $h_A = 0$

λ : $h_\lambda = 1/2$, $h_D = 2$

$\hookrightarrow \lambda$ also has a non-vanishing chiral R-symmetry weight

$c_\lambda = 1/2$ in the Gauge-fixed WZ-multiplet

* The Wess Zumino Gauge multiplet:

$$V_{WZ}^A = \theta \sigma^\mu \bar{\theta} A_\mu^A + i(\theta \theta)(\bar{\theta} \bar{\lambda})^A - i(\bar{\theta} \bar{\theta}) \theta \lambda^A + \frac{1}{2} \theta \theta \bar{\theta} \bar{\theta} D^A$$

$\Rightarrow$ Simply take $P_R \lambda^A := \bar{\lambda}$ to be the spinor field.

* Real multiplet: Vector multiplet which is not Gauge fixed

$(C, P_L, \mathcal{H}, B_a, P_R \lambda, D)$

$\rightarrow B_a$: Poincaré $M^{(0|1)}$ elements.

* Superconformal Gauge multiplet:

In the fixed WZ-gauge, the Super-Action is given by the "super" field strength

$$W_\alpha := -\frac{1}{4} \bar{D}_\alpha \bar{D}^{\dot{\alpha}} D_\alpha V_{WZ}$$

$$\Leftrightarrow \int d^4x \cdot \frac{1}{2} \text{Tr}(F_{\mu\nu} F^{\mu\nu}) \Leftrightarrow \int d^4x \int d^2\theta d^2\bar{\theta} \delta^2(\bar{\theta}) W^\alpha W_\alpha$$

In the WZ-gauge:

$$W_\alpha = -i\lambda_\alpha + \theta_\alpha D + i(\sigma^{\mu\nu} \theta)_\alpha F_{\mu\nu} + \theta \theta (\sigma^\mu \partial_\mu \bar{\lambda})_\alpha$$

$$\therefore \int d^4x W^\alpha W_\alpha = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - 2i\lambda \sigma^\mu \partial_\mu \bar{\lambda} + D^2 + \frac{i}{4} \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma}$$

(For $U(1)$ -Gauge group).

More generally: Take $\partial_\mu \mapsto D_\mu$, $F \in \Omega^2(M) \mapsto \in \mathfrak{g} \otimes \mathbb{R}^2$
(up to gauge terms)

$$S = \int d^4x \text{Tr} \left[-\frac{1}{2} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2} \bar{\lambda} \gamma^\mu D_\mu \lambda + \frac{1}{2} D^2 \right]$$

Note, $\lambda \in S^2(M^0)$ is now $\rightarrow \lambda \in S^2(M^1)$

$$\delta A_\mu^A = -\frac{1}{2} \bar{\epsilon} \gamma_\mu \lambda^A$$

$$\delta \lambda^A = \left[\frac{1}{4} \gamma^{\mu\nu} F_{\mu\nu}^A + \frac{1}{2} i \gamma_\mu D^\mu \lambda^A \right] \epsilon$$

$$\delta D = \frac{1}{2} i \bar{\epsilon} \gamma_\mu \gamma^\mu D_\mu \lambda^A$$

$$D_\mu \lambda^A := \partial_\mu \lambda^A + \lambda^C \lambda_\mu^B f_{BC}^A$$

$$(n+1) \text{ chiral mults: } \Phi^I, i=0, \dots, n \quad \left[\text{(matter-coupled)} \right]$$

$$\Phi^I = X^I(\eta) + \theta \Omega^I(\eta) + \theta \theta F^I(\psi)$$

• One of these is a "compensating mult."

Physical chiral: $(Z^\alpha, \chi^\alpha, F^\alpha), \alpha=1, \dots, n$

WZ Gauge Mult:

$$\delta A_\mu^A = -\frac{1}{2} \bar{\epsilon} \gamma_\mu \lambda^A$$

$$\delta \lambda^A = \frac{3}{2} [\lambda_0 + i \gamma_\mu \lambda^\mu] \lambda^A + \left[\frac{1}{4} \gamma^{\mu\nu} \hat{F}_{\mu\nu}^A + \frac{1}{2} i \gamma_\mu D^\mu \lambda^A \right] \epsilon$$

$$\delta D^A = 2 \lambda_0 D^A + \frac{1}{2} i \bar{\epsilon} \gamma_\mu \gamma^\mu D_\mu \lambda^A$$

or

$$\hat{F}_{ab}^A = e_a^\mu e_b^\nu (2 \partial_{[\mu} A_{\nu]}^A + f_{BC}^A A_\mu^B A_\nu^C + \bar{\psi}_{[\mu} \gamma_{\nu]} \lambda^A)$$

$$D_\mu \lambda^A = \left(\partial_\mu + \frac{3}{2} b_\mu + \frac{1}{4} \omega_{\mu}^{\alpha\beta} f_{\alpha\beta}^A - \frac{3}{2} i \gamma_\mu A_\mu \right) \lambda^A + \lambda^C A_\mu^B f_{BC}^A - \left[\frac{1}{4} \right] \psi_\mu$$

Real-Multiplets: Super Conformal trans: (ϵ, η) ^{only}

$$\begin{aligned} \delta C &= \frac{i}{2} (\bar{\epsilon}_L - \bar{\epsilon}_R) \} \\ &= \frac{i}{2} (\bar{\epsilon} P_L - \bar{\epsilon} P_R) \} = \frac{i}{4} \bar{\epsilon} (\frac{1}{2} + \gamma_5 - (1 - \gamma_5)) \} \\ &= \frac{i}{2} \bar{\epsilon} \gamma_5 \} \end{aligned}$$

$$\delta \zeta_L = \frac{1}{2} \mathcal{H} \epsilon_L - \frac{1}{2} \mathcal{D} \epsilon_R - \frac{1}{2} \mathcal{H} \epsilon_L + \frac{1}{2} \mathcal{D} \epsilon_L$$

$$(\alpha=0)$$

$$\gamma^\mu P_R B_\mu \epsilon = P_L \gamma^\mu B_\mu \epsilon$$

$$\mathcal{D} \epsilon_R = P_L \mathcal{D} \epsilon$$

$$P_L \mathcal{H}$$

$$10$$

$$\delta \zeta_L = \frac{1}{2} P_L (\mathcal{H} - \mathcal{D} \cdot i \mathcal{D}) \epsilon - i \mathcal{H} \epsilon_L \checkmark$$

$$\delta B_a = \left(-\left(\frac{1}{2} \bar{\epsilon}_L D_a \zeta_L\right) - \frac{1}{2} \bar{\epsilon}_L \gamma_a \mathcal{D} \epsilon_R + \frac{1}{2} (1+h) \bar{\eta}_L \gamma_a \zeta_R \right) + \left(-\frac{1}{2} D_a \bar{\zeta}_R \epsilon_R - \frac{1}{2} \bar{\lambda}_L \gamma_a \epsilon_R + \frac{1}{2} (h+1) \bar{\zeta}_L \gamma_a \bar{\eta}_R \right)$$

$$= -\frac{1}{2} (\bar{\epsilon}_L D_a \zeta_L + D_a \bar{\zeta}_R \epsilon_R) + \dots$$

$$= -\frac{1}{2} (\bar{\epsilon} P_L (D_a \zeta) + (D_a \bar{\zeta}) P_R \epsilon)$$

$$\rightarrow D_a \bar{\zeta} P_R \epsilon = - \bar{\epsilon} P_R D_a \zeta$$

$$\rightarrow -\frac{1}{2} (\bar{\epsilon} (\frac{1}{2} (1+\gamma_a)) = \frac{1}{2} (1-\gamma_a))$$

$$= -\frac{1}{2} \epsilon \gamma_a D_a \zeta$$

?

...

Real:

$$\delta C = \frac{1}{2} i \bar{\epsilon} \gamma_a \zeta$$

$$\delta P_L \zeta = \frac{1}{2} P_L (\mathcal{H} - \mathcal{D} - i \mathcal{D} C) \epsilon - i \mathcal{H} \epsilon P_L \mathcal{H}$$

$$\delta \mathcal{H} = -i \bar{\epsilon} P_R (\lambda + \mathcal{D} \zeta) + (h-2) i \bar{\eta} P_L \zeta$$

$$\delta B_a = -\frac{1}{2} \bar{\epsilon} (\gamma_a \lambda + D_a \zeta) + \frac{1}{2} (1+h) \bar{\eta} \gamma_a \zeta$$

$$\delta P_R \lambda = P_R \left[\frac{1}{4} \gamma^{ab} F_{ab} - i \frac{1}{2} D \right] \epsilon + \frac{1}{2} h P_R [\mathcal{H} - \mathcal{D} + i \mathcal{D} C] \mathcal{H} - h X_k P_L \zeta$$

$$\delta D = \frac{1}{2} i \bar{\epsilon} \gamma_a \mathcal{D} \lambda + i h \bar{\eta} (\lambda + \frac{1}{2} \mathcal{D} \zeta) - 2 h \lambda_a D_a C$$

where, $F_{ab} := 2 D_{[a} B_{b]} + \epsilon_{abcd} D^c D^d C$

In superspace, this is $(D_a D_b \epsilon^{ab}) V$

Real multiplets

↓ (WZ - Gauge fix)

Vector Multiplet

- B_m : As a Gauge field, carries conf. weight 0. Thus

$$B_c = e_a^m B_m : \text{Conf. weight } \underline{1}$$

$$w_{B_c} = 1$$

Since $e_a^m = (e^{-1})^m_a \leftarrow \text{weight } \underline{-1}$

Thus, the scalar field C must have weight $w_C = 0$

→ This simplifies many of the susy transf.^s above.

[The following text is a dense, continuous block of characters and symbols, appearing to be a corrupted or heavily redacted document. It contains no legible words or phrases.]